

are contravariant cohomological functors, hence give rise to convergent spectral sequences

$$E_1^{pq} = h^q(L^{-p}) \Rightarrow h^{p+q}(L).$$

One has a morphism of spectral sequences

$$E(\text{for } h') \longrightarrow E(\text{for } h'')$$

which is an isomorphism for $q \leq 0$, and both E^{pq} vanish for $p < 0$ or p large. It follows that $h^n(L) \simeq h'^n(L)$ for $n \leq 0$, i.e. that L satisfies (*).

The same argument can be expressed as an induction on the number of i such that $L^i \neq 0$. If n is the largest (with $n \leq 0$), the induction assumption applies to $\sigma^{<n}L$, even to $(\sigma^{<n}L)[-1]$, and one concludes by the long exact sequence defined by

$$0 \longrightarrow L^n[-n] \longrightarrow L \longrightarrow \sigma^{<n}L \longrightarrow 0.$$

- (c) Expressing L as the inductive limit of the $\sigma^{\geq -n}L$ and using Proposition 11, one sees that we need not assume that L is bounded.
- (d) If $L' \rightarrow L''$ is a quasi-isomorphism, $L' \otimes \mathbb{Z}(\mathbb{A}^1) \rightarrow L'' \otimes \mathbb{Z}(\mathbb{A}^1)$ is one too (flatness of $\mathbb{Z}(\mathbb{A}^1)$), and (*) holds for L' if and only if it holds for L'' .
- (e) Any abelian sheaf L is a quotient of a direct sum of sheaves $\mathbb{Z}(U)$. For instance, the sum over (U, s) , $s \in \Gamma(U, L)$, of $\mathbb{Z}(U)$ mapping to L by s . It follows that L admits a resolution L^* by such sheaves. By (a) and (c), L^* satisfies (*). It follows from (d) that L satisfies (*), and then by (c) that any complex in degree ≤ 0 satisfies (*).

2.4 Application to Presheaves with Transfers

Let F be a presheaf with transfers. A formal argument [7] shows that the presheaves with transfers $H^i C_*(F)$ are homotopy invariant. By the basic result (Theorem 1) recalled in the first lecture, it follows that for any U , one has

$$\mathbf{H}^*(U, C_*(F)) = \mathbf{H}^*(U \times \mathbb{A}^1, C_*(F)) \quad (4.1)$$

Indeed, as U and $U \times \mathbb{A}^1$ are of finite cohomological dimension, both sides are abutment of a convergent spectral sequence

$$E_2^{pq} = H^p(U, H^q C_*(F)) \Rightarrow H^{p+q}(U, C_*(F))$$

and the same for $U \times \mathbb{A}^1$. By Theorem 1 applied to $H^q C_*(F)$,

$$H^p(X, H^q C_*(F)) := H^p(X, aH^q C_*(F))$$

is the same for $X = U$ or for $X = U \times \mathbb{A}^1$. Applying (3.5), we conclude from Proposition 4 (2) $\Rightarrow$ (1) that

Proposition 5. *For k perfect, if F is a presheaf with transfers, for all p , $K(C_*(F)[p])$ is $\mathbb{A}^1$ -local.*

Combining Proposition 5 with Proposition 7 we get the second equality in (1.1).

2.5 End of the Proof of Theorem 2

For any pointed simplicial sheaf $G_\bullet$, $C_\bullet(G_\bullet)$ is a pointed bisimplicial sheaf of which one can take the diagonal $\Delta C_\bullet(G_\bullet)$. For any pointed sheaf G , one has a natural map $G \rightarrow C_\bullet(G)$, and for a pointed simplicial sheaf $G_\bullet$, those maps for the G_n induce

$$a : G_\bullet \longrightarrow \Delta C_\bullet(G_\bullet).$$

Proposition 6. *The morphism $a : G_\bullet \rightarrow \Delta C_\bullet(G_\bullet)$ is an $\mathbb{A}^1$ -equivalence.*

Proof. We deduce Proposition 6 from Proposition 20.

The two maps $0, 1 : F_\bullet \rightarrow F_\bullet \wedge \mathbb{A}_+^1$ are equalized by $F_\bullet \wedge \mathbb{A}_+^1 \rightarrow F_\bullet$, hence become equal in the $\mathbb{A}^1$ -homotopy category. If two maps of pointed simplicial sheaves $F_\bullet \rightrightarrows G_\bullet$ factor as $F_\bullet \rightarrow F_\bullet \wedge \mathbb{A}_+^1 \rightarrow G_\bullet$, they also become equal. By the adjunction of $\wedge \mathbb{A}_+^1$ and of $C_1(-) = \text{Hom}(\mathbb{A}_+^1, -)$, such a factorization can be rewritten as

$$F_\bullet \longrightarrow C_1(G_\bullet) \longrightarrow G_\bullet.$$

Particular case: the maps $C_1(G_\bullet) \rightarrow G_\bullet$ become equal in the homotopy category. Evaluated on X , these maps are the restriction maps $0^*, 1^* : G_\bullet(X \times \mathbb{A}^1) \rightarrow G_\bullet(X)$.

The affine space $\mathbb{A}^n$ is homotopic to a point in the sense that

$$H : \mathbb{A}^1 \times \mathbb{A}^n \longrightarrow \mathbb{A}^n, \quad (t, x) \longmapsto tx$$

interpolates between the identity map (for $t = 1$) and the constant map 0 (for $t = 0$). The map H induces

$$G_\bullet(S \times \mathbb{A}^n) \longrightarrow G_\bullet(S \times \mathbb{A}^n \times \mathbb{A}^1)$$

and, composing with $0, 1$ in $\mathbb{A}^1$, we obtain that

$$G_\bullet(S \times \mathbb{A}^n) \rightrightarrows G_\bullet(S \times \mathbb{A}^n)$$

the identity map, and the map induced by $0 : \mathbb{A}^n \rightarrow \mathbb{A}^n$, are equal in the $\mathbb{A}^1$ -homotopy category. The map of simplicial sheaves $G_\bullet \rightarrow C_n G_\bullet$ is hence an $\mathbb{A}^1$ -equivalence. It has as inverse in the $\mathbb{A}^1$ -homotopy category the map induced by $0 : \text{Spec}(k) \rightarrow \mathbb{A}^n$, and one applies Remark 1.

We now apply Proposition 20 to the bisimplicial sheaves

$$G_{pq} := G_p, \quad H_{pq} := C_q G_p : S \longmapsto G_p(S \times \Delta^q)$$

and to the natural map $G_{pq} \rightarrow H_{pq}$. For fixed q , this is just $G(S) \rightarrow G(S \times \mathbb{A}^q)$, and Proposition 20 gives Proposition 6. $\square$

To prove the last equality in (1.1), it suffices to show that:

Lemma 1. *For any abelian sheaf F , $F[p] \rightarrow C_*(F)[p]$ induces an $\mathbb{A}^1$ -equivalence from $K(F[p])$ to $K(C_*(F)[p])$.*

Proof. For G a monoid (with unit), the pointed simplicial set $B_\bullet G$ is given by

$$B_n G = \left\{ \begin{array}{l} \text{functors from the ordered set } (0, \dots, n) \text{ viewed as a category} \\ \text{to } G \text{ viewed as a category with one object} \end{array} \right\}.$$

This construction can be sheaffied, and can be applied termwise to a simplicial sheaf of monoids, leading to a pointed bisimplicial sheaf of which one can take the diagonal

$$BG_\bullet := \Delta B_\bullet(G_\bullet).$$

This construction commutes with the construction $G_\bullet \rightarrow \Delta C_\bullet(G_\bullet)$. Indeed, $B_n G_p$ is naturally isomorphic to G_p^n ; the operation C_m commutes with products, and $B(\Delta C_\bullet(G_\bullet))$ and $\Delta C_\bullet(BG_\bullet)$ are both diagonals of the trisimplicial pointed sheaf $C_\bullet B_\bullet G_\bullet$.

For abelian simplicial sheaves, the operation B gives again abelian simplicial sheaves, hence can be iterated, and $\Delta C_\bullet$ commutes with B^n .

Via Dold–Puppe construction, B corresponds, up to homotopy, to the shift $[1]$ of complexes:

$$NBG_\bullet \simeq (NG_\bullet)[1].$$

This can be viewed as an application of the Eilenberg–Zilber Theorem (see [9, Theorem 8.5.1]): one has

$$NBG_\bullet \simeq BG_* \simeq \text{Tot } B_* G_* \quad (\text{Eilenberg–Zilber}),$$

and for each G_q , the normalization of $B_\bullet G_q$ is just $G_q[1]$, so that the double complexes $B_* G_*$ and $H_{pq} := G_q$ for $p = 1, 0$ otherwise, have homotopic Tot.

If $G_\bullet$ is an abelian simplicial sheaf, applying Proposition 6 to $B^p G_\bullet$, we obtain that

$$B^p G_\bullet \longrightarrow \Delta C_\bullet B^p G_\bullet = B^p \Delta C_\bullet G_\bullet \quad (6.1)$$

is an $\mathbb{A}^1$ -equivalence. The functor K transforms chain homotopy equivalences into simplicial equivalences. For any simplicial abelian group $L_\bullet$ (to be $G_\bullet$ or $\Delta C_\bullet G_\bullet$), we hence have a simplicial homotopy equivalence

$$B^p L_\bullet = K N B^p L_\bullet \simeq K((NL_\bullet)[p]).$$

Simplicial homotopy equivalences being $\mathbb{A}^1$ -equivalences, we conclude that (6.1) induces an $\mathbb{A}^1$ -equivalence

$$K((NG_\bullet)[p]) \longrightarrow K(N(\Delta C_\bullet G_\bullet)[p]).$$

□

2.6 Appendix: Localization

Let C be a category and S be a set of morphisms of C . The localized category $C[S^{-1}]$ is deduced from C by “formally inverting all $s \in S$ ”. With this definition, it is clear that one has a natural functor

$$\text{loc} : C \longrightarrow C[S^{-1}],$$

bijjective on the set of objects, and that for any category D ,

$$F \longmapsto F \circ \text{loc} : \text{Hom}(C[S^{-1}], D) \longrightarrow \text{Hom}(C, D)$$

is a bijection from $\text{Hom}(C[S^{-1}], D)$ to the set of functors from C to D transforming morphisms in S into isomorphisms.

If one remembers that the categories form a 2-category, and if one agrees with the principle that one should not try to define a category more precisely than up to equivalence (unique up to unique isomorphism), the universal property of $C[S^{-1}]$ given above is doubly unsatisfactory. The easily checked and useful universal property is the following: $F \mapsto F \circ \text{loc}$ is an equivalence from the category $\text{Hom}(C[S^{-1}], D)$ to the full subcategory of $\text{Hom}(C, D)$ consisting of the functors F which map S to isomorphisms.

Proposition 7. *If Y in C is such that the functor*

$$h_Y : C^{op} \longrightarrow \text{Sets} : X \longmapsto \text{Hom}_C(X, Y)$$

transforms maps in S into bijections, then

$$\text{Hom}_C(X, Y) \xrightarrow{\sim} \text{Hom}_{C[S^{-1}]}(X, Y).$$

Proof. By Yoneda construction $Y \mapsto h_Y$, C embeds into the category $C^\wedge$ of contravariant functors from C to Sets , while $C[S^{-1}]$ embeds into $C[S^{-1}]^\wedge$, identified by (a) with the full subcategory of $C^\wedge$ consisting of the F transforming S into bijections. For Y in C , with image $\bar{Y}$ in $C[S^{-1}]$, and for any F in $C[S^{-1}]^\wedge \subset C^\wedge$, one has

$$\text{Hom}(h_Y, F) = \text{Hom}(h_{\bar{Y}}, F).$$

Indeed, by (a) and the Yoneda lemma for C and $C[S^{-1}]$, both sides are $F(Y)$. This means that $h_{\bar{Y}}$ is the solution of the universal problem of mapping h_Y into an object of $C[S^{-1}]^\wedge \subset C^\wedge$. In particular, for Y as in (b), i.e. in $C[S^{-1}]^\wedge$, $h_{\bar{Y}}$ coincides with h_Y , as claimed by (b). $\square$

Proposition 8. *Let (L, R) be a pair of adjoint functors between categories C and D . Let S and T be sets of morphisms in C and D . Assume that F maps S to T and that G maps T to S . Then the functors $\bar{L}, \bar{R}$ between $C[S^{-1}]$ and $D[T^{-1}]$ induced by L and R again form an adjoint pair.*

Proof. The functors $\bar{L}$ and $\bar{R}$ induced by F and G are characterized by commutative diagrams

$$\begin{array}{ccc} C & \xrightarrow{L} & D \\ \downarrow & & \downarrow \\ C[S^{-1}] & \xrightarrow{\bar{L}} & D[T^{-1}] \end{array} \quad \begin{array}{ccc} D & \xrightarrow{R} & C \\ \downarrow & & \downarrow \\ D[T^{-1}] & \xrightarrow{\bar{R}} & C[S^{-1}]. \end{array}$$

Adjunction can be expressed by the data of $\varepsilon : Id \rightarrow RL$ and $\eta : LR \rightarrow Id$, such that the compositions

$$R \longrightarrow RLR \longrightarrow R, \quad L \longrightarrow LRL \longrightarrow L$$

are the identity automorphisms of R and L , respectively (see, e.g., [6]).

By the universal property of localization, ε induces a morphism $\bar{\varepsilon} \rightarrow \bar{R}\bar{L}$, indeed to define such a morphism amounts to defining a morphism $\text{loc} \rightarrow \bar{R}\bar{L}\text{loc}$, and $\bar{R}\bar{L}\text{loc} = \text{loc}RL$. Similarly, η induces $\bar{\eta} : \bar{L}\bar{R} \rightarrow Id$. The morphism $\bar{R} \rightarrow \bar{R}\bar{L}\bar{R} \rightarrow \bar{R}$, and similarly for $\bar{L}$, is induced by $R \rightarrow RLR \rightarrow R$, and the proposition follows. $\square$

Proposition 9. *Suppose that:*

1. *The localization $C[S^{-1}]$ gives rise to a right calculus of fractions.*
2. *Coproducts exist in C , and S is stable by coproducts.*

Then a coproduct in C is also a coproduct in $C[S^{-1}]$.

For the definition of “gives rise to a right calculus of fractions” see [10]. It implies that for X in C , the category of $s : X' \rightarrow X$ with $s \in S$ is filtering, and that

$$\text{Hom}_{C[S^{-1}]}(X, Y) = \text{colim}_{s : X' \rightarrow X} \text{Hom}_C(X', Y).$$

Proof. For X in C , let (S/X) be the filtering category of morphisms $X' \rightarrow X$ in S . For X the coproduct of X_α , $\alpha \in A$, one has a functor “coproduct”

$$\prod_{\alpha \in A} (S/X_\alpha) \longrightarrow (S/X).$$

It is cofinal: for $s : X' \rightarrow X$ in S , one can construct a diagram

$$\begin{array}{ccc} X'_\alpha & \longrightarrow & X_\alpha \\ s_\alpha \downarrow & & \downarrow \\ X' & \xrightarrow{s} & X \end{array}$$

with $s_\alpha : X'_\alpha \rightarrow X$ in S , and $\coprod s_\alpha$ dominates s . For any Y , it follows that

$$\begin{aligned} \mathrm{Hom}_{C[S^{-1}]}(X, Y) &= \mathrm{colim}_{(S/X)} \mathrm{Hom}_C(X', Y) \\ &= \mathrm{colim}_{\prod (S/X_\alpha)} \mathrm{Hom}\left(\prod X'_\alpha, Y\right) \\ &= \mathrm{colim}_{\prod (S/X_\alpha)} \prod \mathrm{Hom}(X'_\alpha, Y) \\ &= \prod_{S/X_\alpha} \mathrm{colim} \mathrm{Hom}(X'_\alpha, Y) = \prod \mathrm{Hom}_{C[S^{-1}]}(X_\alpha, Y), \end{aligned}$$

meaning that X is also the coproduct of the X_α in $C[S^{-1}]$. $\square$

Corollary 10. *Suppose that in the abelian category A arbitrary direct sums exist and are exact. Then arbitrary direct sums exist in the derived category $D(A)$, and the localization functor*

$$C(A) \longrightarrow D(A)$$

commutes with direct sums.

Proof. The functor $C(A) \rightarrow D(A)$ factors through the category $K(A)$ of complexes and maps up to homotopy. Direct sums in $C(A)$ are also direct sums in $K(A)$. Indeed,

$$\begin{aligned} \mathrm{Hom}_{K(A)}(\oplus K_\alpha, L) &= H^0 \mathrm{Hom}^\bullet(\oplus K_\alpha, L) \\ &= H^0 \prod \mathrm{Hom}^\bullet(K_\alpha, L) = \prod H^0 \mathrm{Hom}^\bullet(K_\alpha, L), \end{aligned}$$

as $\prod$ is exact for abelian groups. Exactness of $\oplus$ in A ensures that a direct sum of quasi-isomorphisms is again a quasi-isomorphism, and Proposition 9 applies to $K(A)$ and the set S of quasi-isomorphisms, proving the corollary. $\square$

If A_i , $i \geq 0$, is an inductive system of objects of A , the colimit of A_i is the cokernel

$$\oplus A_i \xrightarrow{d} \oplus A_i \longrightarrow \mathrm{colim} A_i \longrightarrow 0$$

of the difference of the identity map and of the sum of the $A_i \rightarrow A_{i+1}$. If taking the inductive limit of a sequence is an exact functor, the map d is injective: it is the colimit of the

$$\oplus_{i=0}^n A_i \longrightarrow \oplus_{i=0}^{n+1} A_i,$$

each of which is injective, as its graded for the filtration by the $\oplus_{i \geq p} A_i$ is the identity inclusion.

Under the assumptions of Corollary 10 if a complex K is the colimit of an inductive sequence $K_{(i)}$, and if the sequence

$$0 \longrightarrow \oplus K_{(i)} \xrightarrow{d} \oplus K_{(i)} \longrightarrow K \longrightarrow 0 \quad (10.1)$$

is exact, then for any L , the long exact sequence of cohomology reads

$$\longrightarrow \mathrm{Hom}(K, L) \longrightarrow \prod \mathrm{Hom}(K_{(i)}, L) \xrightarrow{d} \prod \mathrm{Hom}(K_{(i)}, L) \longrightarrow$$

The kernel of d is simply the projective limit of the $\mathrm{Hom}(K_{(i)}, L)$. The cokernel is $\lim^1$. One concludes.

Proposition 11. *Suppose that in A countable direct sums exist and are exact. If the complex K is the colimit of the $K_{(i)}$, and if the sequence (10.1) is exact, for instance if either:*

1. *in A inductive limits of sequences are exact;*
2. *in each degree n , each $K_{(i)}^n \rightarrow K_{(i+1)}^n$ is the inclusion of a direct factor;*

then one has a short exact sequence

$$0 \longrightarrow \lim^1 \mathrm{Hom}(K_{(i)}, L[-1]) \longrightarrow \mathrm{Hom}(K, L) \longrightarrow \lim \mathrm{Hom}(K_{(i)}, L) \longrightarrow 0.$$

Proof. It remains to check that condition (2) implies the exactness of (10.1). This is to be seen degree by degree. By assumption, the $A_i := K_{(i)}^n$ have decompositions compatible with the transition maps $A_i = \bigoplus_{j=0}^i B_j$. A corresponding decomposition of (10.1) in direct sum follows, and we are reduced to check exactness of the particular case $(10.1)_{(A,n)}$ of (10.1) when $B_i = 0$ for $i \neq n$, i.e. when A_i is a fixed A from $i \geq n$, and is 0 otherwise. Let $(10.1)_0$ be the sequence $(10.1)_{\mathbb{Z},n}$ in Ab for $A = \mathbb{Z}$. It is a split exact sequence of free abelian groups. Because direct sums exist, $L \otimes A$, for L a free abelian group, is defined and functorial in L . It is a sum of copies of A , indexed by a basis of L , and is characterized by

$$\mathrm{Hom}(L \otimes A, B) = \mathrm{Hom}(L, \mathrm{Hom}(A, B))$$

(functorial in B). The sequence $(10.1)_{A,n}$ is $(10.1)_0 \otimes A$, and, $(10.1)_0$ being split exact, it splits and in particular is exact. $\square$

The truncation $\sigma_{\leq n} K = \sigma_{\geq -n} K$ of a complex K is the subcomplex which coincides with K in homological degree $\leq n$ and is 0 in homological degree $> n$. For any complex K , one has

$$K = \operatorname{colim}_{\sigma_{\leq n}} K$$

and this colimit satisfies condition (2) of Proposition 11. It follows that

Corollary 12. *Under the assumptions of Corollary 10, for any K and L , one has a short exact sequence*

$$0 \longrightarrow \lim^1 \mathrm{Hom}(\sigma_{\leq n} K, L[-1]) \longrightarrow \mathrm{Hom}(K, L) \longrightarrow \lim \mathrm{Hom}(\sigma_{\leq n} K, L) \longrightarrow 0.$$

3 $\mathbb{A}^1$ -Equivalences of Simplicial Sheaves on G -Schemes

3.1 Sheaves on a Site of G -Schemes

We fix a base scheme S , supposed to be separated noetherian and of finite dimension; fiber product $X \times_S Y$ will be written simply as $X \times Y$. We also fix a group scheme G over S , supposed to be finite and flat. We note h_X the representable sheaf defined by X .

Let QP/G be the category of schemes quasi-projective over S , given with an action of G . Any X in QP/G admits an open covering (U_i) by affine open subschemes which are G -stable. This

makes it possible to define a reasonable quotient X/G in the category of schemes over S (rather than in the larger category of algebraic spaces). For each U_i , U_i/G is defined as the spectrum of the equalizer

$$\mathcal{O}(U_i) \rightrightarrows \mathcal{O}(U_i \times G),$$

and X/G is obtained by gluing the U_i/G . It is a categorical quotient, i.e. the coequalizer of $G \times X \rightrightarrows X$. The map $X \rightarrow X/G$ is finite, open, and the topological space $|X/G|$ is the coequalizer of the map of topological spaces $|G \times X| \rightrightarrows |X|$. One can show that X/G is again quasi-projective. Remark 2 below shows that this fact, while convenient for the exposition, is irrelevant.

One defines on QP/G a pretopology [3, II.1.3] by taking as coverings the family of étale maps $Y_i \rightarrow X$ with the following property: X admits a filtration by closed equivariant subschemes $\emptyset = X_n \subset \cdots \subset X_1 \subset X_0 = X$ such that for each j , some map $Y_i \rightarrow X$ has a section over $X_j - X_{j+1}$. The Nisnevich topology on QP/G is the topology generated by this pretopology. The category QP/G with the Nisnevich topology is the *Nisnevich site* $(QP/G)_{\text{Nis}}$.

Remark 1. *The corresponding topos is not the classifying topos of [2, IV.2.5]. A morphism $X \rightarrow Y$ can become a Nisnevich covering after forgetting the action of G , and not be a Nisnevich covering. Example: $S = \text{Spec}(k)$, $G = \mathbb{Z}/2$, $X = S$, $Y = S \amalg S$, and G permutes two copies of S in Y .*

Remark 2. *Let $(\text{affine}/G)_{\text{Nis}}$ be the site defined as above, with “quasi-projective” replaced by “affine”. It is equivalent to $(QP/G)_{\text{Nis}}$, in the sense that restriction to $(\text{affine}/G)_{\text{Nis}}$ is an equivalence from the category of sheaves on $(QP/G)_{\text{Nis}}$ to the category of sheaves on $(\text{affine}/G)_{\text{Nis}}$.*

Remark 3. *If G is the trivial group e , the definition given above recovers the usual Nisnevich topology. For $G = e$, the condition usually considered: “every point x of X is the image of a point with the same residue field of some Y_i ”, is indeed equivalent to the condition imposed above. This is checked by noetherian induction: if a generic point ξ of X can be lifted to Y_i , some open neighborhood $U \subset X$ of ξ can be lifted to Y_i , and one applies the induction hypothesis to $X_1 = (X - U)_{\text{red}}$.*

We write $(QP)_{\text{Nis}}$ for the category of quasi-projective schemes over S , with the Nisnevich topology.

Lemma 2. *If $U : Y_i \rightarrow X$, $i \in I$, is a covering of X in $(QP/G)_{\text{Nis}}$, there is a covering $\mathcal{V}$ of X/G in $(QP)_{\text{Nis}}$ whose pull-back to X is finer than U .*

Proof. Fix a filtration $\emptyset = X_n \subset \cdots \subset X_0 = X$ as in the definition of the Nisnevich topology. We write q for the quotient map $X \rightarrow X/G$. For x in X/G , $(q^{-1}(x))_{\text{red}}$ is in some $X_j - X_{j+1}$, by equivariance of the X_j , and one of the maps $Y_i \rightarrow X$ has an equivariant section s over $X_j - X_{j+1}$. Let $(X/G)_x^h$ be the henselization of X/G at x . The map q being finite, the pull-back of $(X/G)_x^h$ to X is the coproduct of the X_y^h for $q(y) = x$. The map from Y_i to X being étale, the section s , restricted to $(q^{-1}(y))_{\text{red}}$, extends uniquely to a section (automatically equivariant) of Y_i over $\amalg_{q(y)=x} X_y^h$. Writing $(X/G)_x^h$ as the limit of étale neighborhoods of x , one finds that x has an étale neighborhood $V(x)$ such that Y_i has an equivariant section over $X \times_{X/G} V(x)$. The $V(x)$ form the required covering $\mathcal{V}$. $\square$

We define the G -local henselian schemes to be the schemes Y obtained in the following way. For X in (QP/G) , y a point of X/G , and $(X/G)_y^h$ the henselization of X/G at y , take the fiber product

$$Y := X \times_{X/G} (X/G)_y^h.$$

As X is finite over X/G , this fiber product is a finite disjoint union of local henselian schemes, and G -local henselian schemes are simply the G -equivariant finite disjoint unions of Y of local henselian schemes, for which Y/G is local.

Proposition 13. *If Y is G -local henselian, the functor $X \mapsto \text{Hom}(Y, X)$ is a point of the site $(QP/G)_{\text{Nis}}$, i.e. it defines a morphism of the punctual site (Sets) to $(QP/G)_{\text{Nis}}$. If $Y = X \times_{X/G} (X/G)_y^h$, the corresponding fiber functor is*

$$F \mapsto \varinjlim_{Y \subset X \times_{X/G} V} F(X \times_{X/G} V),$$

the colimit being taken over the etale neighborhoods of y in X/G . The collection of fiber functors so obtained is conservative.

Proof. The functor $X \mapsto \text{Hom}(Y, X)$ commutes with finite limits. It follows from Lemma 2 that it transforms coverings into surjective families of maps, hence is a morphism of sites $(\text{Sets}) \rightarrow (QP/G)_{\text{Nis}}$.

To check that the resulting set of fiber functors is conservative, it suffices to check that a family of etale $f_i : U_i \rightarrow X$ is a covering if for any G -local henselian Y ,

$$\coprod_i \text{Hom}(Y, U_i) \longrightarrow \text{Hom}(Y, X)$$

is onto. The proof, parallel to that of Lemma 2, is left to the reader. $\square$

3.2 The Brown–Gersten Closed Model Structure on Simplicial Sheaves on G -Schemes

We recall that a commutative square of simplicial sets (or pointed simplicial sets)

$$\begin{array}{ccc} K & \longrightarrow & L \\ \downarrow & & \downarrow \\ M & \longrightarrow & N \end{array} \quad (13.1)$$

is homotopy cartesian (or a homotopy pull-back square) if, when L is replaced by L' weakly equivalent to it and mapping to N by a (Kan) fibration $L \xrightarrow{\sim} L' \rightarrow N$, the map from K to $L' \times_N M$ is a weak equivalence.

Definition 3. *A simplicial presheaf $F_\bullet$ on $(QP/G)_{\text{Nis}}$ is flasque if $F(\emptyset)$ is contractible and if for any (upper) distinguished square*

$$\begin{array}{ccc} B & \longrightarrow & Y \\ \downarrow & & \downarrow p \\ A & \xrightarrow{j} & X \end{array}$$

(p etale, j open embedding, $B = p^{-1}(A)$, and $Y - B \simeq X - A$), the square

$$\begin{array}{ccc} F(X) & \longrightarrow & F(Y) \\ \downarrow & & \downarrow p \\ F(A) & \xrightarrow{j} & F(B) \end{array}$$

is homotopy cartesian.

Theorem 3. *Let $f : F_\bullet \rightarrow F'_\bullet$ be a morphism of flasque simplicial presheaves. If the induced morphism of simplicial sheaves $aF_\bullet \rightarrow aF'_\bullet$ is a local equivalence, then, for any U in QP/G , $F_\bullet(U) \rightarrow F'_\bullet(U)$ is a weak equivalence.*